

t -Probing (In-)Security

Pitfalls on Noise Assumptions

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Abstract. The ongoing transition to post-quantum cryptography has led to a surge of research in side-channel countermeasures tailored to these schemes. A prominent method to prove security in the context of side-channel analysis is the utilization of the well-established t -probing model. However, recent studies by Hermelink et al. at CCS 2024 demonstrate a simple and practical attack on a provably secure implementation of the Fujisaki-Okamoto transform.

In this paper, we present an unsupervised single-trace side-channel attack on a tenth-order masked implementation of fixed-weight polynomial sampling. Notably, this masking scheme has also been proven to be secure in the t -probing model. Both attacks reveal a mismatch between the correct, well-understood theory of the t -probing model and its practical application – the security proofs are valid, yet the attacks still succeed at high noise levels. Therefore, we take a closer look at the underlying causes and the assumptions made for transferring t -probing security to practice. In particular, we investigate the amount of noise required for this transfer. We find that, depending on the design decisions, the additional noise required can be substantial and difficult to achieve.

Consequently, we examine the factors that impact the required amount of additional noise and which need to be considered for practically secure implementations. In particular, non-uniformly distributed shares – a setting that is increasingly encountered in post-quantum cryptographic algorithms – can lead to an increased noise requirement, and thus could reduce the security level of the masked implementation. Our analysis allows us to provide practical guidelines for masking scheme designers, thereby facilitating the development of theoretically *and* practically secure designs.

Keywords: t -probing model, SCA, PQC, Masking, Mutual Information, Noise

1 Introduction

Since the late nineties, Side-Channel Analysis (SCA) has been known as a major threat to cryptographic implementations. By observing the (physical) behavior of the target device, such as runtime differences [Koc96], power consumption [KJJ99], or electromagnetic emission [GMO01] during secret-dependent computations, the adversary can learn the processed secret. Potential targets include smartcards, key fobs, and even smartphones, all of

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which run cryptographic services under (physical) exposure to the adversary. Consequently, the scientific community has developed countermeasures, of which masking [CJRR99] is most notable. The core idea is to split secret intermediate values and related operations into multiple shares such that all subsets of shares are independent of the secret. As side-channel leakage is correlated with intermediate values of the implementation, masking of sensitive (secret or secret-dependent) values reduces this dependency and, thus, their observable leakage. In order to evaluate the efficacy and validity of masked operations, a plethora of methods ranging from empirical leakage testing and formal models have been proposed in the past e.g., [SM15, PR13, DDF14].

One popular formal analysis method to reason about side-channel resistance is the t -probing model, introduced by Ishai, Sahai and Wagner [ISW03]. This model provides an abstraction of the adversarial capabilities, as it allows the (modeled) adversary to learn up to t intermediate values of the computation. To ensure security in the t -probing model, every set of intermediate variables with cardinality of up to t must be statistically independent of the secret. The noisy leakage model [PR13], on the other hand, captures real-world leakage more closely by modeling the noisy side-channel observations as probabilistic functions of intermediate variables, but it is therefore more difficult to work with. To connect these models, Duc et al. [DDF14] propose a reduction from the noisy leakage to the t -probing model that also shows under which assumptions t -probing security translates into practical security. As a result, specific requirements on the amount of noise of masked implementations emerge which must be met for t -probing security to translate into practice.

One highly relevant domain for the application of the t -probing model is Post-Quantum Cryptography (PQC), where a multitude of new masking schemes have been introduced and proven secure recently [BPO⁺20, BGR⁺21, KLRBG22, CGTZ23]. This relevance stems from the possible advent of powerful quantum computers in the next decade(s), which forces the scientific, industrial, and regulative communities, e.g., the National Institute of Standards and Technology (NIST) [NIS17], to focus on PQC as quantum-resistant alternative. This effort already resulted in the first set of standards [oST24a, oST24b, oST24c]. Consequently, the migration and integration of these standards into digital systems is currently ongoing. Here, the t -probing model is seen as a valuable cornerstone to establish sound countermeasures against SCA.

Just recently, however, Hermelink et al. demonstrated a full-key recovery attack against a proven t -probing secure implementation with relatively low effort [HNPS24]. In particular, they target the masked implementation of the Fujisaki-Okamoto (FO) transform by D’Anvers et al. [DBV23] to recover the key of the NIST standard ML-KEM [oST24a] (formerly Kyber [SAB⁺20]). While the theoretical proof in the t -probing model is correct and promises side-channel security, the simplicity and practicality of the attack point toward a misalignment in the application of the t -probing model when implementing for practical security. The FO transform [FO99] is a widespread component in PQC used to transform Chosen-Plaintext Attack (CPA) secure decryption into Chosen-Ciphertext Attack (CCA) secure decapsulation.

Another component that is deployed in many PQC schemes is Fixed-Weight Polynomial Sampling (FWPS), which samples a random polynomial with a fixed weight. Due to its relevance, for instance in HQC [AAB⁺22] and NTRU [CDH⁺20], it has been the target of vigorous research with respect to side-channel countermeasures. One efficient way to achieve a t -probing secure solution for FWPS is to use shuffling and to construct a masked permutation as a combination of shares, where each share is a random permutation [CGTZ23]. Yet, in this work, we show a simple and practical attack against an implementation following this approach. Again, the theoretical proof of the scheme in the t -probing model is correct. The explanation for the feasibility of the attack is that the amount of noise required for the masking scheme to also be secure in practice is, in this case, not attained. This highlights a mismatch between the theoretical understanding

and practical application of t -probing security that increasingly occurs in the context of PQC. Although it is well known that noise is required for masking to provide security in practice [CJRR99, SVO⁺10], this aspect is sometimes neglected in recently presented masked algorithms and implementations.

Contributions. In this paper, we discuss and specify the noise conditions required to transfer the guarantees of t -probing security into reality. In particular, we find that under certain conditions, the requirements on the amount of noise are difficult to meet in practice and discuss how they might be avoided by design choices. Our contributions are threefold:

First, we highlight that the noise requirements for translating t -probing security into practical security have recently been disregarded (Section 3). We demonstrate this by attacking a higher-order masked provable t -probing secure implementation of FWPS. In particular, we show an unsupervised attack on a *tenth-order* masked implementation and illustrate that the presented attack becomes even simpler with increasing masking order. One key factor enabling this attack is a violation of the requirement of a uniform sharing, which in turn increases the noise requirement for practical security.

Second, we investigate the effect of non-uniformity and its implications for security (Section 4). While the reduction of the noisy leakage model to the t -probing model assumes that the sensitive values (e.g., masking shares) are uniformly distributed, this is not always the case in practice. We demonstrate that the amount of required noise, and, thus, the real-world security varies significantly when the sensitive values are not uniformly distributed leading to potential vulnerabilities. We then show that in our examples the dispersion index quantifies the impact of non-uniformity and provide a straightforward method to evaluate the effect of non-uniformity in masked implementations.

Third, we derive three main factors that further influence the required amount of additional noise of t -probing-secure schemes (Section 5). We analyze their impact on the leaked information and relate our findings to prior results. Our analysis provides an intuitive method to understand and assess how the implementation inherent noise behaves and to avoid devastating mistakes in t -probing secure implementations. These findings allow us to derive practical guidelines for the development of t -probing secure implementations. Further, we apply our approach to our novel attack in Section 3 and the attack of [HNPS24] to give a conceptual explanation on how these vulnerabilities arise.

Related Work. A well researched limitation of the t -probing model, first stated by Battistello et al. [BCPZ16], is the missing coverage of horizontal attacks. In their work, they use the reduction of Duc et al. [DDF14] which implies that the t -probing proof can only be extended to the noisy leakage model, and thus, to practice, if the noise of the implementation increases with masking order t . After considering the t -probing model, Battistello et al. [BCPZ16] describe the effect of a horizontal attack on the required noise for a secure implementation. More precisely, they show that an n -time repeated processing of the same value leads to an averaging of the normal distributed noise. This results in a decrease of the implementation noise by a factor of $\sqrt{n}$.

Due to this and other limitations, Duc et al. [DFS19] propose a different approach to demonstrate the security of masking schemes. Instead of applying a security model, they use the Mutual Information (MI) in order to derive lower bounds for the security level of a masked implementation. However, as for t -probing security, these bounds only hold in practice for a "sufficiently large" noise level. Furthermore, they discuss that the sufficient amount of noise depends on the leakage function. The problem of insufficient noise is also addressed by Masure et al. [MMMS23]. They show that arithmetic masking leaks substantially less information than Boolean masking in low noise settings (in Hamming Weight (HW) and bit leakage model) and propose a new masking scheme that uses small-Mersenne-prime encodings for sharing secret values. This allows them to protect implementations on low-noise devices without loss of security.

These works assume uniformity of sensitive values. In contrast, Masure and Standardaert [MS23] not only emphasize that the "measure of noisiness of the leakage" is part of security proofs in general. They also discuss in Theorem 2 [MS23] the effect of the non-uniformity of sensitive values, stating that the information leakage of a non-uniformly distributed sensitive values could be increased by $\alpha \cdot |\mathcal{Y}|$, where α is a constant factor and $|\mathcal{Y}|$ is the field size. Nonetheless, this upper bound does not necessarily provide concrete guidance to practitioners, and it is therefore the purpose of this paper to fill this gap by analyzing the noise factors that affect the practical security of the *t*-probing model.

2 Preliminaries

In this paper, we use the term "sensitive values" to refer to the secret dependent intermediate values that are attacked in SCAs. In a masked implementation, this refers to the shares.

2.1 Side-Channel Analysis

Side-channel attacks exploit unintended information leakage, such as variations in power consumption [KJJ99] or timing [Koc96], to extract secrets from a cryptographic implementation. In the context of power and Electromagnetic (EM) side-channel attacks, a distinction is made between single-trace and multi-trace attacks. This classification depends on whether the attacker can observe only one or many instances of the implementation using the same secret. In multi-trace attacks, the attacker gathers multiple measurements over several executions, enhancing the potential for revealing the secret by analyzing the relationship between different traces. In power SCA, a typical hardware behavior is that the power consumption correlates with the number of bits set to one, often referred to as HW leakage [MOP07]. In contrast, value leakage occurs when the information leakage directly depends on the data values processed by the circuit.

Noise in Electronic Circuits. However, since many other electronic components and non-secret data contribute to variations in the power consumption, the measurable signal associated with the sensitive value is often obscured by noise. This noise is influenced by factors of the measurement setup and the device, such as low bandwidth of the connection between the logic cells and the measuring oscilloscope, as well as the clock frequency and hardware implementation effects [MOP07]. Nonetheless, in most electronic circuits, noise tends to follow a normal distribution [MOP07]. Thus, it makes it more difficult to extract meaningful data from the power measurements and to protect the implementation from side-channel attacks. In the following, the implementation-inherent noise is referred to as the noise level of the implementation.

2.2 Masking

The implementation of side-channel countermeasures such as masking can complement this device- and setup-specific noise [CJRR99]. Masking is a well-established countermeasure against SCAs, such as power or EM attacks, that follows the secret sharing approach introduced by Shamir [Sha79]. The basic idea behind masking is to split secret values into uniform random shares. This ensures that the leakage observed by an adversary does not depend directly on the secret, as the attacker is unable to link the observed data to the actual value of the secret. Usually, the number of shares is denoted by $t + 1$. By increasing the number of shares, the complexity of the attack grows exponentially, as the attacker must analyze multiple leakage values to infer the secret. In effect, masking amplifies the noise in the system, which significantly hinders the attacker's ability to extract the secret [LBBS20]. The effectiveness of masking lies in its ability to hide the secret within

random noise, increasing the difficulty for attackers to exploit side-channel information and extract sensitive data. It is implemented at the algorithmic level, this making it a versatile and efficiently developable countermeasure.

2.3 Information Theory

Information theory provides tools to systematically reason about leakage obtained in side-channel attacks and thus, to estimate the security level of a masking scheme. In the following, we give a short overview on commonly used quantities in SCA and refer to [MOP07] for a more thorough introduction to the subject.

Signal-to-Noise Ratio. The Signal-to-Noise Ratio (SNR) is a commonly used metric to relate the amount of useful information – the *signal*, often a sensitive value y – to the noise (see, e.g., [MOP07]). It is defined as the ratio of the variances of the signal and the noise. In the context of SCA, the SNR expresses how much the leakage value y drawn from a probability distribution, denoted as Y , vary in comparison to the noise.

Entropy and Differential Entropy. The SNR relates the signal to the noise, but it does not capture the information that a variable holds. Instead, the *uncertainty* of a discrete random variable X is given by its entropy [Sha48], which is defined as

$$H[X] = - \sum_{x \in \mathcal{X}} P(X = x) \log_2(P(X = x)), \quad (1)$$

where $\mathcal{X}$ is the domain of X , and we set $P(X = x) \log_2(P(X = x)) = \lim_{p \rightarrow 0} p \log_2(p) = 0$ for $P(X = x) = 0$. Clearly, $H[X]$ is always positive and is maximized by the uniform distribution over the respective support. If X is continuous, one possible generalization is differential entropy. For a random variable X with density function p , it is defined as

$$H[X] = - \int_{\mathcal{X}} p(x) \log_2(p(x)) dx. \quad (2)$$

Unfortunately, differential entropy is not necessarily positive and reaches its maximum if X follows a normal distribution. In the following, we use integrals to express both sums and integrals; note that this can be formalized using measure theory where a sum can be expressed as integration with respect to the counting measure.

The entropy of a random variable conditioned on another variable is particularly important in the area of SCA. If we, for example, obtain leakage L on a secret key X , we may be interested in how much uncertainty, i.e., how much entropy, is left in the secret *under the knowledge of L* . This quantity is expressed by conditional entropy. For a discrete random variable Y , one can see the conditional entropy as weighted average of $H(X \mid Y = y)$ over all elements of the domain of Y , $\mathcal{Y}$; concretely, we have that

$$H[X \mid Y] = \sum_{l \in \mathcal{Y}} P(Y = l) H[X \mid Y = l]. \quad (3)$$

Mutual Information. Given two random variables X and L , we are interested in how much information we learn on X given L , i.e., how much the entropy of X is reduced given L . For a discrete random variable, this reduction in entropy is called *mutual information* [Sha48] and is defined as

$$I(X; L) = H[X] - H[X \mid L] = H[L] - H[L \mid X] = H[X] + H[Y] - H[X, Y]. \quad (4)$$

For continuous random variables, we replace entropy by differential entropy. Mutual information is 0 if X and L are independent and, otherwise, positive but smaller or equal than both $H(X)$ and $H(Y)$. In many settings, the reduction of entropy is of interest only in relation to the total entropy, e.g., of a secret; therefore, it makes sense to normalize by the entropy of the secret. We denote the normalized mutual information by

$$NI(X; L) = \frac{H(X) - H(X | L)}{H(X)}. \quad (5)$$

In the discrete case, the normalized mutual information is strictly bounded by 1. However, for continuous random variables, the differential entropy can be negative. Therefore, the normalized mutual information can be > 1 in the continuous case.

2.4 Probing Models

To simplify the analysis of specific implementations and systematically analyze the conditions for security, formal security models for SCA have been introduced. Here, abstract definitions of the system behavior and the adversarial capabilities are utilized to prove specific security properties. A limitation from any formal security proof is that its validity is tightly bound to its specific model. This can only be mapped to real-world conditions when ensuring the assumptions are met – which is a non-trivial task in general. Two widely used formal models that aim to capture the specific context of SCA, are the *noisy-leakage model* [PR13] and the *t-probing model* [ISW03].

Noisy-Leakage Model. In the (serial) noisy-leakage model [PR13], the adversary Adv_δ gets access to a noisy function of each intermediate variable. That is, for each intermediate y , with ℓ being the number of intermediate variables, the leakage is resembled by a uniformly distributed random variable L_y that is a function of y_i and a random noise variable $\mathbf{N}$. Here, the noise is selected such that the bias of the intermediate variable given L_i is bounded by δ . A common, but not model-inherent, assumption in the literature is the application of additive Gaussian noise [BCO04, SVO⁺10, DFS15, MOP07], i.e.,

$$L_y = \mathbf{f}(y) + \mathbf{N} = \mathbf{f}(y) + \mathcal{N}(0, \sigma^2), \quad (6)$$

where $\mathbf{f}$ is a deterministic function and $\mathbf{N}$ is drawn from a normal distribution (the model requires that the noisyness of L_y is upper bounded in some metric [BCGR24]). Then, instead of considering the bias given by L_y , we can equivalently define the noise level over the variance σ^2 of the normal distribution (as this is the only factor that determines the bias). An important assumption in the noisy-leakage model is that each L_y leaks independent of each other, i.e., for each $i, j \in [\ell]$ we have that $\mathbf{N}_i$ and $\mathbf{N}_j$ are statistically independent. Since the noisy-leakage model was originally proposed for the analysis of uniform masking schemes [PR13, Definition 2], another assumption made is that each masking value occurs with the same probability.

t -Probing Model. While the noisy-leakage model is close to real-world leakage, it is quite complex to directly work with. Hence, simpler models were developed, of which the t -probing model [ISW03] is certainly the most widespread. In this model, the adversary Adv_t can freely select up to t intermediate values and learn the exact values. Hence, the adversary selects a set of probes $\mathcal{P} \subset [\ell]$, with $|\mathcal{P}| \leq t$, and the leakage is then defined by the set $\mathcal{L}_{\mathcal{P}} = \{y_i \mid i \in \mathcal{P}\}$. Let $L_{\mathcal{P}}$ be a random variable defined over the set $\mathcal{L}_{\mathcal{P}}$. We then say, an implementation is t -probing secure if the set $\mathcal{L}$ is always statistically independent of the secret input $\mathbf{S}$, i.e., for all $\mathcal{P}$ of cardinality at most t we have $\Pr[L_{\mathcal{P}} \mid \mathbf{S}] = \Pr[L_{\mathcal{P}}]$.

In contrast to the noisy-leakage model, the t -probing model does not capture the uncertainty of the adversary when observing side-channel leakage. Instead, the adversary

gets all information but of a restricted subset of intermediates. The advantage of the t -probing model comes from its simplicity and the fact that (under certain noise conditions) security in the noisy-leakage model can be reduced to t -probing security [DDF14].

Reduction between Models. It has been shown, that the view of the noisy-leakage adversary Adv_δ , with $\delta = d/(4\ell \cdot |\mathcal{Y}|)$ for some $d > 0$ and computations over the set $\mathcal{Y}$, can be simulated by the probing adversary Adv_t , when $t = \frac{d}{2} - 1$, with probability $p_{\text{sim}} \geq 1 - e^{-d/12}$ [DDF14]. Hence, with probability p_{sim} the adversary in the noisy-leakage model cannot learn more than t intermediate values. The core idea of the proof involves an interleaved model, where the adversary gets each intermediate value y_i with some probability p (usually called the p -random probing model [ISW03]). The reduction from the noisy-leakage to the p -random probing model works quite well, because here the noise is still incorporated in the leakage of each intermediate value, by the probability p . The reduction from the p -random probing to the t -probing model then basically computes the probability that at most t of the intermediate values are indeed given to the adversary. Hence, the noisy observation of the original adversary gets lost in the probability p_{sim} . Please note, due to the reduction, the assumptions required for the noisy-leakage model transfer also to the t -probing model. In particular, the assumption of uniformly distributed masking is (implicitly) involved by conditioning the bias δ on the set $\mathcal{Y}$ over which the masking is happening.

3 Revisiting t -Probing Security in Applications

Since the introduction of the t -probing model in 2003 [ISW03], it has proven to be a useful tool for analyzing the resistance of masked cryptographic implementations to power side-channel attacks. The t -probing model often allows for simple proofs, and has been used for many masking schemes. This includes proofs for countermeasures applied to PQC as shown in e.g., [dPKPR24, MGTF19, BPO⁺20]. However, the attack we present in the following, shows that the noise requirements for the t -probing model vary vastly in practice and must be considered closely. To this end, we demonstrate a successful unsupervised attack against the t -probing secure FWPS masking scheme by Coron et al. [CGTZ23], and discuss the occurring discrepancy between the theoretical model and the practical implementation. In order to show that our attack is not an isolated incident applying to FWPS, but a more principled issue in the application of t -probing secure schemes, we discuss the similarities between our attack and the recent attacks on masked Kyber by Wang et al. [WBD24] and Hermelink et al. [HNPS24].

3.1 Secure Sampling of a Fixed-Weight Polynomial

Many PQC cryptosystems require secret polynomials to be sampled from the set of N -bit strings with weight W , i.e., from $\mathcal{S}_{N,W} = \{\mathbf{v} \in \{0,1\}^N : \sum_{i=0}^{N-1} v_i = W\}$, with $N, W \in \mathbb{N}^+, W < N$. Examples include the key generation of NTRU [CDH⁺20] as well as the decapsulation of the designated NIST standard algorithm HQC [AAB⁺22].

A prominent way to sample a fixed-weight polynomial is *shuffling*. For this approach, the vector is initialized with a fixed polynomial of correct weight W , and shuffled afterwards (i.e., a random permutation is applied) to obtain a uniform random polynomial with the correct weight. A common method to receive a random permutation of the initialized vector is to apply sorting. Here, a random integer is associated to each vector element v_i , and these tuples are sorted according to the random numbers. Often sorting networks are used due to their timing side-channel security and efficiency. Thus, the core operations of sorting are the comparison of integers and – depending on its outcome – a swap of the two tuples. In software, these swaps are usually implemented by using swap masks to prevent

Algorithm 1 FWPS Masking Scheme proposed by Coron et al. [CGTZ23].

```

1:  $g_0 := (\underbrace{1, \dots, 1}_{W/2}, \underbrace{-1, \dots, -1}_{W/2}, \underbrace{0, \dots, 0}_{N-W})$ 
2:  $g_i := (\underbrace{0, \dots, 0}_N)$  for  $1 \leq i \leq t$ 
3: for  $i := 0$  to  $t$  do
4:   draw random permutation  $\pi_i$ 
5:   for  $j := 0$  to  $t$  do
6:      $g_j := \pi_i(g_j)$ 
7:   refresh sharing  $g_0, \dots, g_t$ 
8: return  $g_0, \dots, g_t$ 

```

timing side-channels. So, for a swap, all bits of the mask are set to one (mask = $2^{n_{\text{bits}}} - 1$). In contrast, if the tuples are not swapped, the mask is set to zero (mask = 0).

Masked Implementation. Coron et al. propose a masking scheme for the FWPS of NTRU [CGTZ23] which divides the random permutation into several sequential shares as shown in Algorithm 1. First, the first share of the secret is initialized as a vector g with the correct weight W , and all other shares are initialized to zero. Subsequently, a random permutation is applied to each share independently. Importantly, the permutation is the same for each share. After this, the sharing is refreshed in order to restore the independence between the shares itself and the previous shared data. These two steps – repeated permutation and refreshing – are iterated $t + 1$ times in total. It is trivial to see that this holds secure in the t -probing model: by definition, the adversary can only probe t permutations while information about one permutation is missing. Conveniently, this results in a much more efficient construction than applying Boolean masking at a lower level as done by Krausz et al. [KLRBG23]. Note that the algorithm and its proof work on a high level of abstraction, and thus allow for any random permutation function π_i .

3.2 Our Attack on [CGTZ23]

Our practical attack against the masked permutation by Coron et al. [CGTZ23] works on a much lower abstraction level than the proof. Furthermore, as our attack is unsupervised, it does not require profiling or manual threshold calibration. In the following, we present the algorithmic idea of the attack, before we discuss the practical aspects and the noise requirements of the reduction from the t -probing model.

Algorithmic Level. For NTRU, the target in [CGTZ23], the sampling is performed in the key generation. Thus, a real-world attack must work by only measuring one single power trace of one execution. This execution consists of the computation of $t + 1$ sorting-based permutations. For each of those sorting operations, we denote the number of performed swaps by n_{swaps} . n_{swaps} is approximately $n \cdot \log(n)$, for NTRU (with $n = 509$) and in our case $n_{\text{swaps}} \approx 4500$. Since we attack a software implementation, the swaps are executed sequentially, and the information about all swap decisions are included in one power trace of one single execution. The resulting attack strategy is shown in Algorithm 2. The initial vector is public and always the same. Note that in the specific case of NTRU, the sampling target set has $W/2$ ones and $W/2$ minus ones. Consequently, the sampling procedure can be adapted trivially. Hence, one possible way to recover the secret (masked) key polynomial g , is to identify all swap masks and thus trace the complete $t + 1$ permutations.

Therefore, recovering each swap mask correctly, yields the exact final key. For this

Algorithm 2 Attack on Coron et al. [CGTZ23].

```

1: initialize  $g$ 
2:  $g \leftarrow (\underbrace{1, \dots, 1}_{W/2}, \underbrace{-1, \dots, -1}_{W/2}, \underbrace{0, \dots, 0}_{N-W})$ 
3: measure power trace of one execution
4: divide power trace into  $t + 1$  permutation traces
5: for  $i := 0$  to  $t$  do ▷ iterate over permutation traces
6:   divide permutation trace  $i$  into  $n_{\text{swaps}}$  parts
7:   for  $j := 0$  to  $n_{\text{swaps}} - 1$  do ▷ iterate over swaps
8:     for  $k := 0$  to  $t$  do ▷ iterate over swap shares
9:       recover swap hypothesis  $\text{hyp}_k \in \{0, 1\}$ 
10:    majority vote over  $\text{hyp}_0, \dots, \text{hyp}_t$ 
11:    conditional swap within  $g$  based on majority vote result
12: return  $g$ 

```

purpose, we decide to use a majority voting on the $t + 1$ executions of each swap to extract the corresponding swap mask. Even if this method is not optimal, it suffices for an attack that succeeds with higher probability for a higher masking order, which we discuss in more detail later. After recovering all the swap masks, it is possible to redo the permutations on the initial vector with the recovered swap masks and retrieve the final key. Thus, no further side-channel information but the measurement of the swaps is necessary to fully recover the key. The success rate of this attack highly depends on the masking order and on the number of swaps in one permutation. The latter results from the fact that the probability for recovering the entire key is equal to the product of all probabilities for recovering the individual swap masks. Even for a very high swap-recover rate of 99 %, this naive attack methodology only results in an overall success probability of $0.99^{n_{\text{swaps}}}$. However, the fact that many vector elements are the same, helps the attack to succeed also for lower swap recovery probabilities, since some swaps are meaningless after all.

Practical Aspects. The hardware platform used for the practical attack is the Chip-Whisperer¹ framework equipped with a STM32F303RCT6 ARM target featuring a 32-bit ARM Cortex-M4 architecture. The power trace is measured over a 12Ω shunt resistor by an external PicoScope 6402B oscilloscope with a resolution of 8 bits and a sample rate of 7812500 samples/s. Afterwards, the measured trace is divided into $n_{\text{swaps}} \cdot (t + 1)$ subtraces, each covering an execution of one swap. For each subtrace, a hypothesis for each corresponding swap share is made, based on a simple threshold. Then, the $t + 1$ hypotheses of the swap shares are used to do a majority vote to find the best hypothesis for the swap. This procedure is done for all swaps. No pre- or postprocessing of the measured traces or the guessed swap masks are required. In the following, we discuss two effects that make this attack possible.

Decreasing Security with Increasing Masking Order. Analyzing the masking scheme proposed by Coron et al. [CGTZ23] reveals that each permutation is applied $t + 1$ times on the different data shares. However, the permutations on the shares are the same for all shares. Hence, by combining the $t + 1$ horizontal results to one, the recovery probability of one swap is increased. The masking leads to a repetition that works in this specific case as an *amplifier* of the (secret) signal and, thus, increases the SNR. This supports the findings of Battistello et al. [BCPZ16], who showed that an n -time repeated processing of the same value leads to a decrease of the implementation noise by a factor of $\sqrt{n}$. At first glance,

¹<https://rtfm.newae.com/Targets/CW303%20Arm/>

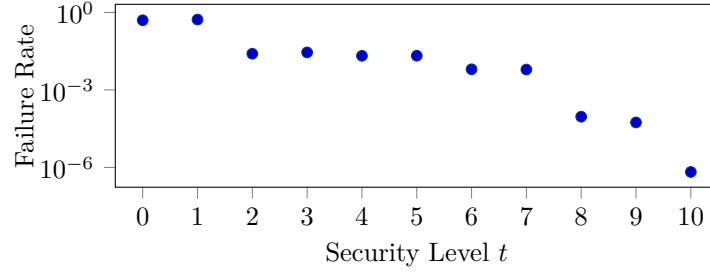


Figure 1: Swap recovery failure rate of the masking by Coron et al. [CGTZ23] at different security levels t measured over 500 000 swaps.

it seems counter-intuitive that an increasing masking order would increase the required noise level. However, this is partly explained by the reduction of the noisy leakage to the t -probing model. In particular, to get an exponential increase in security, we require both a linear increase in the masking order and noise [DDF14]. In our case, the noise initially remains constant with a higher masking order. Additionally, it is then further reduced by a factor of $\sqrt{n}$ due to the effect of the horizontal attack. So in practice, the gap between the required noise and the real-world noise grows with increasing masking order.

We observe this in practice, which is visualized in Figure 1 showing how effective the masking is (the probability of a wrong swap recovery) depending on the masking order. It can be seen that an increasing masking order in general leads to a decreasing masking efficacy as the correct swap recovery rate increases vastly. However, the stepped shape originates from the attack procedure of the majority vote. This is because the majority-threshold is computed as $\lceil \frac{t}{2} \rceil$. A better decision algorithm could lead to a more efficient usage of the side-channel information obtained in the attack, thus, resulting in a strictly increasing recover probability. By utilizing this effect, we are able to run a successful attack on a 10^{th} order masked implementation with a success rate of 99.8%.

Non-Uniformity. The ability to apply a simple threshold to the data is due to the widespread distribution of the two possible values. In the sorting-shuffling implementation on a 32-bit architecture, they are $y \in \{\text{no swap}, \text{swap}\} = \{0, 32\}$, assuming Hamming weight leakage. The use of only two out of the 33 possible values (with respect to their Hamming weight), along with the large distance between them, drastically reduces the complexity of distinguishing between them. In other words: Even for a high noise level, only a marginal number of measurements are in the range where they are incorrectly recovered. Note that the same argumentation applies analogously to value leakage. Figure 2 illustrates how this shows up in practice, by showing ten measured traces, each of a swap (red) and a no-swap (blue). The difference in leakage between the two values can even be recognized by a mere visual inspection of the traces such that no advanced statistical methods are needed to recover the swap-decision. It would take a much higher noise level to achieve leakage values that are not that easy to distinguish, which we further discuss in Section 4.

Analysis. The question that arises is why the attack on this implementation is possible, even though it has been proven to be secure in the t -probing model. A first attempt to explain this discrepancy is the horizontal aspect of the attack (the $t + 1$ -fold repetition of a swap), since horizontal attacks are known to be feasible on t -probing secure implementations [BCPZ16]. However, the horizontal aspect only improves the attack, but is not mandatory. In fact, the non-uniformity of the leakage values alone enables a successful attack. The explanation is found in the t -probing reduction itself. Masking and the reduction of the t -probing proof to noisy leakage, and thus, real-world leakage, are only

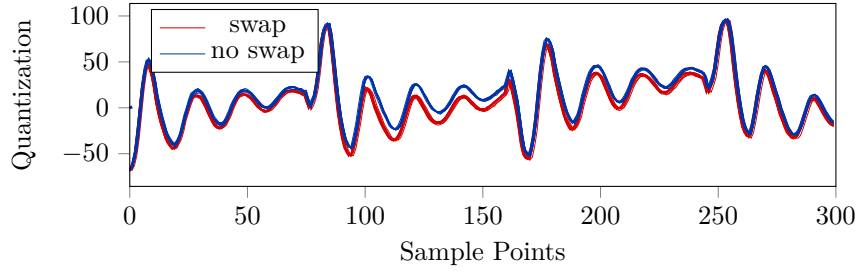


Figure 2: Ten sample traces for a swap operation shown in red and ten sample traces for a non-swap operation shown in blue. The two classes can clearly be distinguished.

defined on uniformly distributed data. Masking in the context of the noisy leakage model is only defined for the uniform distribution [PR13, Definition 2]. This triples down through the reduction to the t -probing model. As discussed before, in our case, the data is not uniformly distributed in the range $[0, 32]$, but can only have two values, 0 and 32, each with probability $P(y) = \frac{1}{2}$. The masked FWPS algorithm by [CGTZ23] can also be realized based on e.g., Fisher-Yates instead of sorting, which would still lead to non-uniformly distributed values, but in a significantly weaker form. Instead of two different values, the attack would have to distinguish N different values with similar or even overlapping Hamming weight. However, the horizontal aspect of the attack applies to all variants of the algorithm by [CGTZ23]. Alternative solutions for masked FWPS as presented by [KLRBG23] do not apply masking at the permutation level and, thus, neither suffer from repetitions of values and non-uniformity, but result in slower implementations. Our attack does not apply to these.

Note that at the algorithmic level, one share corresponds to a uniform random permutation and, thus, does not violate the requirements of the t -probing model. However, due to limitations of the Instruction Set Architecture (ISA), a permutation must be expressed with several intermediate values for a practical implementation. In our case, sorting requires swap masks which can lead to non-uniformity. The same issue is highlighted by two attacks on other PQC masking implementation, which we discuss in the following subsection.

3.3 Kyber

Security of many lattice-based Key Encapsulation Mechanisms (KEMs) first requires INDistinguishability against CPAs (IND-CPA) as foundation of a secure Public-Key Encryption (PKE). From this PKE, INDistinguishability against CCAs (IND-CCA2) secure KEM can be achieved by using the FO-transform: during the decapsulation, the received ciphertext is decrypted and re-encrypted using the same randomness derived from the message; only if both ciphertexts are equal, a shared secret is derived. Specifically, the message bits are encoded into a polynomial by multiplying each bit with $(q+1)/2$ to yield the single coefficients. In software implementations, often a bitmask (e.g., 0x0000 or 0xFFFF) is used so that the computation is constant time. Finally, the submitted ciphertext ct and the re-encrypted ciphertext ct' are compared in order to protect against CCAs.

Previous Attacks on Message Encoding. Wang et al. [WBD24] target the masked message-to-polynomial encoding by Heinz et al. [HKL⁺22] that follows the idea by Oder et al. [OSPG18], which works as follows:

1. Both Boolean shares (single bits) are multiplied separately by $(q+1)/2$.

2. In case both input bits were one, the output masking now would represent $(q+1)/2 + (q+1)/2 = 1 \bmod q$ instead of zero.
3. Consequently, a secure correction is applied for this case to yield zero as well.

This procedure is recalled in [WBD24, Fig. 6]. Since the conversion maps each message bit share *individually* to $(q+1)/2$ if it is one, and to zero if it is zero, it is trivial to see that the function `masked_poly_frommsg()` is t -probing secure with $t = 1$. However, the difference in the leakage of the possible share values is large, because of the two mask values zero vs. `0xFFFF`, and zero vs. $(q+1)/2$. Due to this, Wang et al. [WBD24] were able to reveal both share values (at masking order $t = 1$) and thus, the unshared, unprotected intermediate value.

Moreover, they attack the *higher-order* implementation of the message encoding by Bronchain and Cassiers [BC22] that makes use of the single-bit Boolean-to-arithmetic conversion et al. [SPOG19, Alg. 5]. Again, single masked bits are mapped to a non-uniform, broader distribution that is easier to distinguish in practice. This occurs especially during a key subroutine [SPOG19, Alg. 7] where an existing arithmetic sharing is updated such that the output represents the original value xor an additional bit, which is exploited to perform the overall conversion. In particular, each share a_i of the input sharing is updated as $a_i := a_i - 2(a_i \cdot x) \bmod q$, where x represents the input bit. Hence, computing $a_i \cdot x$ results in easily distinguishable leakages depending on the single bit x .

Both attacks are based on the same effects as we observed in our attack described above: The distribution of the intermediate value (mask) is non-uniform and only features two values with extreme Hamming distance, which leads to a decreased noise level.

Masking of the FO-Transform Comparison [DBV23]. Several works exist that propose masking schemes for the FO transform comparison [OSPG18, BPO⁺20, DHP⁺22, BGR⁺21]. D’Anvers et al. [DBV23] not only stress the insecurity of these previous methods, but also introduce a new t -probing secure scheme. Their method checks whether the q -arithmetically masked ciphertext difference $\Delta \mathbf{ct}$ between $\mathbf{ct}$ and $\mathbf{ct}'$ is zero. Instead of using integer multiplication and arithmetic masking, the authors utilize a comparison over a Galois field. A uniformly random s -bit value R_j is chosen and the sum

$$E_j = \bigoplus_{i=1}^d R_j \otimes \Delta \mathbf{ct}_j^{(i)} \quad (7)$$

is computed, where $\oplus$ and $\otimes$ denote addition and multiplication in $\mathbb{F}_q^s$, respectively. The comparison outputs a decryption failure if and only if

$$\sum_j E_j \not\equiv 0 \pmod{2^{s-1}}. \quad (8)$$

In practice, unless a Galois multiplication instruction is available, the comparison of [DBV23] multiplies each bit of $\Delta \mathbf{ct}_i$ of each share with R_i and xors the results together.

Attack on [DBV23]. Hermelink et al. [HNPS24] show that the masked comparison of [DBV23] is vulnerable to decryption failure attacks. Even though the algorithm is t -probing secure, their attack is still possible at very high noise levels and masking orders. It targets the multiplication of R_j with the shares $b_j^{(i)}$ of the bit b_j . Assuming the noisy HW model, the attacker obtains either a noisy observation of HW 0 or of a random HW of an s -bit integer. These two values are very clearly distinguishable in practice, the leakage behavior of a 0 and a random s -bit integer differ heavily. While the gadget is t -probing secure, each share of each bit can be recovered even under very high noise conditions.

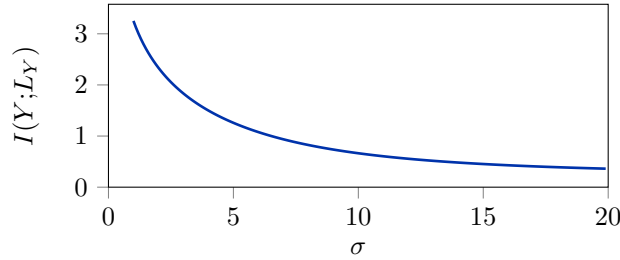


Figure 3: Dependency between the noise level and the mutual information given a uniform distribution (cf. Section 4.2).

Given that, every bit carries information about the secret binary value the adversary is targeting. This gives the attacker a large number of recoverable bits which all may reveal the secret and, thus, the success rate of the attack is vastly increased. Additionally, as in the previously described attacks (ours and Wang et al. [WBD24]), the intermediate values are not uniformly distributed. However, in contrast to these attacks, the cause of the non-uniformity is not a mask, but a multiplication of a single bit with a random value.

It is important to note that this attack is not limited to ML-KEM, but applies to any cryptosystem that deploys the FO-transform and the proposed comparison. As such, it also applies to the designated PQC standard HQC. In the following section, we further investigate the impact of different non-uniform distributions of intermediate values in order to gain a better understanding of the implications on the leakage, and thus on the security of an implementation.

4 The Impact of Non-Uniformity

In the previous section, we demonstrated a successful attack on the t -probing secure FWPS algorithm by Coron et al. [CGTZ23]. The vulnerability stems mostly from the non-uniform distribution of its sensitive values. This finding highlights the importance of understanding how the distribution of sensitive values affects information leakage and, consequently, the side-channel security of cryptographic implementations. As introduced in Section 2.4, the reduction from the t -probing model to the noisy leakage model, which is considerably closer to the real world, assumes uniformly distributed values [PR13, Definition 2]. As a consequence, the reduction does not hold when non-uniformly distributed values are used. In fact, Masure and Standaert show in [MS23] that non-uniformly distributed intermediate values could lead to larger noise requirements. However, for the verification of complex PQC masking techniques, the t -probing model is often used without considering the distribution of the sensitive intermediate values. This raises the question of the concrete practical impact of using non-uniform values in the t -probing model. In order to assess this impact, we use the MI as a detailed measure of the side-channel security as by Duc et al. in [DFS19]. We begin by briefly reviewing the role of MI in side-channel analysis and its connection to the noise level of an implementation. Subsequently, the influence of the probability distributions of both sensitive values and leakage observations on the MI is discussed. Then, we explore how different probability distributions of sensitive values impact the overall MI and identify the conditions that increase or decrease the noise requirements compared to the case of uniformly distributed Y .

4.1 Mutual Information

To understand the effect of missing uniformity of sensitive values, we use the MI that serves as a fundamental metric to quantify the information dependency between sensitive

values Y and observed leakage values L_Y . As Duc et al. state in [DFS19], the MI between an intermediate result and its leakage value can be expressed as

$$I(Y; L_Y) = H[Y] + \sum_{y \in Y} P(y) \cdot \sum_{l_y \in L} P(l_y|y) \log_2 P(y|l_y).$$

Since we assume, as common in the literature e.g., [BCO04, SVO⁺10, DFS15, MOP07], additive Gaussian noise, L_Y is a Gaussian Mixture Model (GMM) with

$$L_Y = \sum_{y \in \mathcal{Y}} P(y) (f(y) + \mathcal{N}(0, \sigma)) = \sum_{y \in \mathcal{Y}} P(y) \mathcal{N}(\mu_Y, \sigma),$$

and we obtain the MI

$$I(Y; L_Y) = H[Y] - \sum_{y \in Y} \left[P(y) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{x - \mu_y}{\sigma}\right)^2\right) \log_2 \left(\frac{\sum_{i \in Y} P(i) \exp\left(-\frac{1}{2} \left(\frac{x - \mu_i}{\sigma}\right)^2\right)}{P(y) \exp\left(-\frac{1}{2} \left(\frac{x - \mu_y}{\sigma}\right)^2\right)} \right) dx \right].$$

This gives us the connection between the MI and the noise level of a circuit. Recall that an increasing noise σ leads to a decreasing MI $I(Y; L_Y)$ which results in an increasing security level. Figure 3 visualizes this effect using an example of a uniform distribution with 32 possible values of Y . As the MI between the secret and the leakage increases, the noise level decreases accordingly, reducing the security level as more information about the secret is revealed through the leakage.

Next, the impact of non-uniformity on the security level of an implementation is investigated. Therefore, in order to obtain the connection between the MI and the leakage L_Y directly, we switch to the equivalent definition of the MI, that is given by

$$I(Y; L_Y) = H[L_Y] - H[L_Y|Y].$$

With the entropy of one element of the GMM, i.e., the $P(Y)$ -weighted sum of Gaussian distributions with mean Y and noise σ , we see that

$$H[L_Y|Y] = \frac{1}{2} \cdot \log_2 (2\pi \cdot \sigma^2)$$

only depends on the noise σ . In other words, it can be seen that the MI only depends on the noise σ and the probability distribution of the leakage L_Y . Unfortunately, the entropy of a GMM cannot be calculated in closed form [HBDH08]. Therefore, we simulate the leakage distributions L_Y for differently distributed Y with a fixed $\sigma = 4$ in the following and use Monte Carlo integration to compute the MI. In order to get results for the MI that we can compare independently of the amount of information in Y , we compute for each case the normalized MI $NI(Y; L_Y)$.

4.2 Baseline: Uniform Distribution

To evaluate the impact of different distributions on the (normalized) MI, we first determine the MI for a uniform distribution of Y with 32 possible values as a baseline. The respective distribution of L_Y is shown in Figure 4. It can be seen that the elements of the GMM blur into another for $\sigma = 4$. This leads to a normalized MI given by $NI(Y; L_Y) = 0.304$.

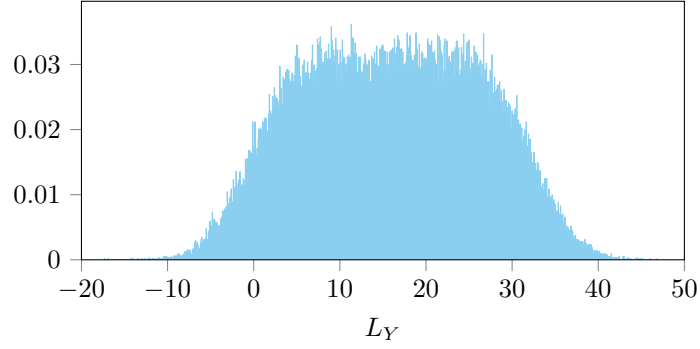
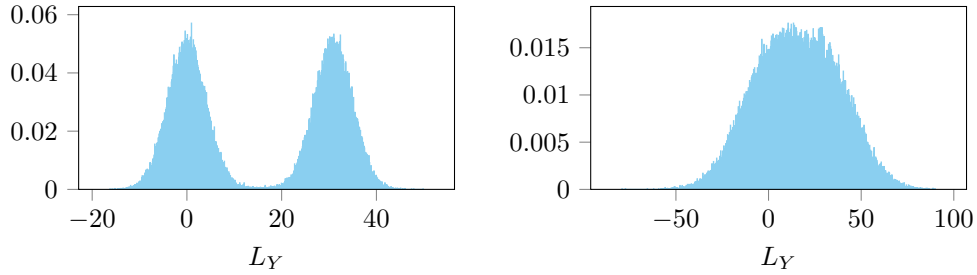


Figure 4: Probability distribution of the leakage value L_Y for a uniform distribution.



(a) Leakage value L_Y of [CGTZ23] assuming $\sigma = 4$. (b) Leakage value L_Y of [CGTZ23] assuming $\sigma = 16$.

Figure 5: Probability distributions for the leakage value L_Y assuming different noise values.

4.3 Binary Distribution

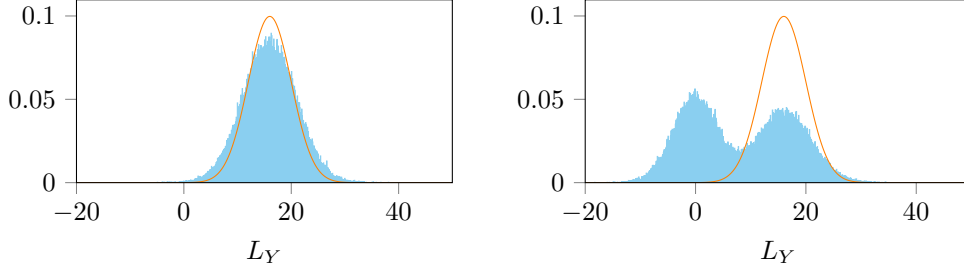
To start with, Figure 5a shows the probability distribution of the physical leakage value L_Y of the practically vulnerable implementation of Coron et al. [CGTZ23]. In the attack, we have $Y \in \{\text{swap}, \text{noswap}\} = \{0, 32\}$. Hence, the GMM in L_Y consists of two elements whose means are rather distant:

$$L_Y = \mathcal{N}(\mu_{\text{swap}}, \sigma) + \mathcal{N}(\mu_{\text{noswap}}, \sigma).$$

Again, we exemplarily choose $\sigma = 4$ to generate Figure 5a. The leakage distribution of L_Y gives us an intuition that it is rather easy to differentiate between the two elements of the GMM, meaning it is straightforward to recover the sensitive values in an SCA. This aligns well with the observations of Wang et al. [WBD24]. Further, this intuition is confirmed by calculating numerically the MI of this implementation (with a fixed $\sigma = 4$). This results in the normalized MI of $NI(Y; L_Y) = 1.113$. The picture changes for $\sigma = 16$, depicted in Figure 5b, where the normalized MI decreases to $NI(Y; L_Y) = 0.611$. The two elements of L_Y are overlapping so that they cannot be distinguished as simple as before. This effect is consistent with Figure 3. The more noise we have in an implementation, the smaller the MI and the higher the security level.

Comparing the normalized MI results for uniform and binary distributed Y shows that a non-uniform² distribution of Y could lead to a higher MI. As a result, the noise level decreases which reduces security. This observation highlights a root cause of the problems with the technique by Coron et al. [CGTZ23] and Heinz et al. [HKL⁺22] in practice:

²We view the binary distribution as non-uniform, because it assigns probability zero to all values between the extremes. The resulting effect is further discussed in Section 5.3.



(a) Probability distribution of the leakage value L_Y for a binomial distribution. (b) Probability distribution of the leakage value L_Y for the FO transform in [DBV23].

Figure 6: Comparison between the leakages of a binomial distribution and the leakage inherent to the FO transform technique by D’Anvers et al. [DBV23].

While the t -probing security requires a much higher noise level for practical security, the binary distribution of the intermediate values drastically reduces the noise level, resulting in a lower level of security for a given additional noise.

4.4 Binomial Distribution

Let us now consider a binomial distribution. In fact, most side-channel attacks assume HW leakage rather than value leakage. With uniform original values, their HWs follow a binomial distribution. Therefore, the distribution of L_Y with a binomial distribution of Y is shown in Figure 6a. As highlighted, the distribution of L_Y barely deviates from that of a normal distribution $\mathcal{N}(\mu, \sigma)$ (given in orange) with μ being the mean of Y . Furthermore, the individual elements of the GMM of L_Y cannot be distinguished visually. The numerical calculation of the normalized MI for $\sigma = 4$ results in $NI(Y; L_Y) = 0.145$. Hence, the MI for binomially distributed Y is significantly smaller than that for uniform values, indicating reduced leakage.

4.5 FO-Transform

In addition to the three presented distributions, there are other, less common distributions. One of them is discussed in the following. Figure 6b shows the distribution of L_Y for the implementation of the FO-transform by D’Anvers et al. [DBV23], as revisited in Section 3.3. Although the data bits are masked with uniformly distributed random bits, the distribution Y of the sensitive values given by the product between the secret and the randomness, is not uniformly distributed. More precisely, the probability for $Y = 0$ is given by $P(Y = 0) = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2^{32}}$, while all other possible values are binomially distributed with a probability given by $P(Y = y, y > 0) = \frac{1}{2} \cdot \frac{1}{2^{32}} \binom{32}{y}$. In Figure 6b, it can be seen that this distribution deviates from a normal distribution (in orange) and, hence, requires more external noise than a uniform distribution. Furthermore, numerically calculated, we get a normalized MI of $NI(Y; L_Y) = 0.444$. Thus, this implementation – also proven t -probing secure – requires more noise to achieve the same security level as a uniform distribution. This explains the success rate at high noise levels and masking orders of the attack of Hermelink et al. [HNPS24] (also explained in Section 3.3) as the result of the multiplication of R_j with $b_j^{(i)}$ is binomially distributed.

To summarize, we highlighted that non-uniformly distributed intermediate values change the noise requirements, and hence, the security level of an implementation, as it is also stated by Masure and Standaert in [MS23]. There, they derive that the MI of a non-uniformly distributed value can be up to a factor of $\alpha \cdot |\mathcal{Y}|$ greater than that of a

uniformly distributed value, where α is a constant factor and $|\mathcal{Y}|$ is the field size of the values. Since this is only an upper bound, it aligns perfectly with our more practical and more detailed results in this section. Our analysis suggests that non-uniform distributions which are closer to normal distributions are more secure, because they require less noise for the same security level. This is also backed by the basic fact that the continuous entropy is maximized by a normal distribution for a fixed σ . On the contrary, those distributions that differ more from the normal distribution have a higher MI and, thus, a lower security level. For developers of new masking schemes, that means if Y has a distribution that is not uniform, the (normalized) MI $NI(Y; L_Y)$ should be calculated in order to check whether more additional noise is needed to achieve the same security level. In the case of $NI(Y; L_Y) \leq NI(Y; L_{\text{uniform}})$, the t -probing model can be applied as usual. On the contrary, if $NI(Y; L_Y) > NI(Y; L_{\text{uniform}})$, the model should be applied with more caution.

4.6 Post-Quantum Cryptography and Non-Uniformity

In Section 3, we exemplarily analyzed two scenarios in which the masked operation is not linear over the respective ring of the PQC scheme (FWPS and FO-Transform). A generic approach is to convert to Boolean masking and subsequently evaluate the circuit that performs the operation. Unfortunately, this generic method is often inefficient, for example if a circuit needs to be evaluated on a single Boolean-masked value, so that bitslicing does not apply efficiently. As a consequence, more efficient proposals have been developed for specific procedures, which have been validated using the t -probing model. However, as just shown, the achieved improvements sometimes come at the cost of violating uniformity.

5 Essential Conditions for Practical t -Probing Security

So far, we have shown that important conditions for applying the t -probing model are sometimes not fulfilled, especially in the context of PQC. Hence, the analysis of the MI seems to be crucial when designing a new masking scheme. Since this method is very expensive, we investigate the MI from a general perspective to identify universal factors that affect the MI and, thus, the noise level of an algorithm. The results of this investigation provide a set of practical guidelines to assist the design of new masking schemes. These guidelines aim to strengthen the practical security level of a t -probing secure implementation, like the sun shining light on current pitfalls in the noise assumptions.

In the following, we first propose a shortcut to analyze the effect of non-uniformity. To this end, we investigate the connection between the variance of a random variable Y representing a sensitive value via the dispersion index to the MI. Next, we recall how the horizontal effect of repeated values exactly influences the noise level of a circuit. Afterwards, we derive the influence of the sensitive value's encoding on the noise level, before the connection between the number of possible probing sets and the noise level is investigated. Eventually, we show why these factors become more important for PQC schemes.

5.1 Variance and Dispersion Index of Sensitive Values

In order to abridge the MI analysis from the previous section, we derive a simple upper bound for the MI. Since a normal distribution maximizes the entropy [CT01], we can estimate the MI as

$$I(Y; L_Y) = H[L_Y] - H[L_Y|Y] \leq H[\mathcal{N}(0, \text{Var}(L_Y))] - H[L_Y|Y]$$

Table 1: Comparison of dispersion index $D(Y)$ and normalized MI $NI(Y; L_Y)$ with $\sigma = 4$.

Distribution of Y	$D(Y)$	$NI(Y; L_Y)$
Binary (swap/non-swap)	15.39	1.113
FO-Transform	8.50	0.444
Uniform	5.67	0.304
Binomial	0.50	0.145

where $H[\mathcal{N}(\mu, \sigma)] = \frac{1}{2} \log_2(2\pi\sigma^2)$. Applying the law of total variance of a GMM, i.e., $\text{Var}(L_Y) = \sigma^2 + \text{Var}(Y)$, we can estimate $I(Y; L_Y)$ as

$$I(Y; L_Y) \leq \log_2 \left(\frac{\text{Var}(L_Y)}{\sigma^2} \right) = \log_2 \left(1 + \frac{\text{Var}(Y)}{\sigma^2} \right). \quad (9)$$

Hence, it is possible to provide an upper bound of the MI of an implementation that depends only on the variance of Y and the noise σ . Therefore, the variance of Y is a practical indicator for the MI, or the required noise level securing a target implementation. This dependency can also be observed when analyzing the SNR. Applying it to Equation 9, we obtain an upper bound for the MI which aligns with [DFS19], i.e.,

$$I(Y; L_Y) \leq \log_2(1 + \text{SNR}(Y, \sigma))$$

In our simulations, we observe (see Table 1) that the dispersion index [Hoe43] $D(Y) = \frac{\text{Var}(Y)}{\mu_Y^2}$ actually serves as a better metric to estimate the effect of the distribution of Y on the noise level. More precisely, we are only interested in the relative “spread” of two distributions of Y without considering the mean value which is provided by the dispersion index. For example, the binomial distribution with $p = 0.5, n = 256$ (a HW leakage of 256 bits) has variance 32, the same as a binary distribution with values $x \in \{0, 16\}$, although the latter is clearly easier to classify for the same noise level. The dispersion index corrects this: for binomial distributions it always equals to $1 - p$ (0.5 in the example), and for the exemplary binary distribution, it yields 8. This intuition allows developers of new masking schemes to easily determine how the non-uniform distribution of Y likely affects the noise level of the implementation.

Noise Factor 1. *The distribution of the sensitive values impacts the noise level. If the values are not distributed uniformly, it might be reduced. A high dispersion index $D(Y) = \frac{\text{Var}(Y)}{\mu_Y^2} > D(Y_{\text{uniform}})$ indicates a high MI and an increased required noise.*

5.2 Repetition of Values

The noise level σ of a leakage value L_Y is reduced by a factor of $\sqrt{n}$, if y is processed n times, as discussed in Section 3.2 and by Battistello et al. [BCPZ16]. More precisely, the noise from the individual measurements averages out in a horizontal attack, and the Gaussian distributions with the same mean μ and variance σ^2 can be combined into one Gaussian distribution: $\mathcal{N}(\mu, \sigma^2/n)$. In the scenario of a side-channel adversary, the leakage of a value that occurs in the same way at n points in time can be simulated by $L_Y = \mathcal{N}(\mu_Y, \sigma^2/n)$. When inserting the merged noise $\sigma' = \sigma/\sqrt{n}$ into the MI, we obtain

$$I(Y; L_Y) = H[Y] - H \left[Y \middle| \mathcal{N} \left(\mu_Y, \frac{\sigma^2}{n} \right) \right]. \quad (10)$$

Figure 7 illustrates this effect on the MI for a uniform distribution of Y and a constant σ . For an implementation designer, this means that the repetition of values should be

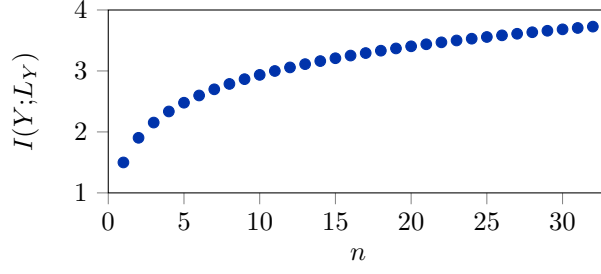


Figure 7: Dependency between the number of repetitions and the MI given a uniform distribution of 32 possible values (cf. Section 4.2).

avoided. If the repetition is strictly necessary, just maintaining the same level of security requires increasing the noise σ by $\sqrt{\max(n)}$. Recall that our attack against the masking scheme of Coron et al. [CGTZ23] becomes easier with an increasing masking order, as we showed in Section 3.2. In particular, each swap-share is repeatedly processed $t + 1$ times and – for instance by a majority vote – these leakage values can be combined to effectively reduce the noise. The straightforward mitigation for avoiding the repeated processing is not possible here as the algorithm specifically *requires* it.

Noise Factor 2. *The n -fold repetition of a sensitive value leads to a $\sqrt{n}$ increase of the required additional noise.*

5.3 Distance of Means

Furthermore, the required noise level is also directly influenced by the implementation and the device on which it is applied. Due to the regular structure of the underlying hardware, we can assume the mean values μ_y of the elements of the GMM of L_Y to be evenly distributed with distance a between two neighbouring values y , i.e., $\mu_y = a \cdot y$. In most cases, these are all values in $[0, 2^l]$, or their corresponding Hamming weights. By integrating the formula of the MI with this definition of the mean values through substitution ($z = \frac{x}{a}$, $dx = a \cdot dz$), we obtain

$$I(Y; L_Y) = H[Y] - \sum_{y \in Y} \left[P(y) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \frac{\sigma}{a}} \exp \left(-\frac{1}{2} \left(\frac{z-y}{\frac{\sigma}{a}} \right)^2 \right) \log_2 \left(\frac{\sum_{i \in Y} P(i) \exp \left(-\frac{1}{2} \left(\frac{z-i}{\frac{\sigma}{a}} \right)^2 \right)}{P(y) \exp \left(-\frac{1}{2} \left(\frac{z-y}{\frac{\sigma}{a}} \right)^2 \right)} \right) dz \right]. \quad (11)$$

Here, we can observe that increasing the distance of means a reduces the security level of the implementation because it affects the MI in the opposite way as the noise. In other words: As the distance of means increases, the noise must also be increased to maintain the same level of security. Figure 8 illustrates this effect by comparing two different scenarios, both with the same $\sigma = 4$, as it shows the distribution of L_Y . It demonstrates that an attacker is better able to differentiate between the two values for the distribution with $a = 10$. This factor is related to the previously discussed impact of the probability distribution of Y , because both, the distribution and the distance of means, influence the distribution of L_Y and, thus, the noise level of the implementation. However, the distance of means is an effect of the encoding of Y and depends on the implementation details and the hardware, while the distribution of Y is a characteristic of the masking scheme.

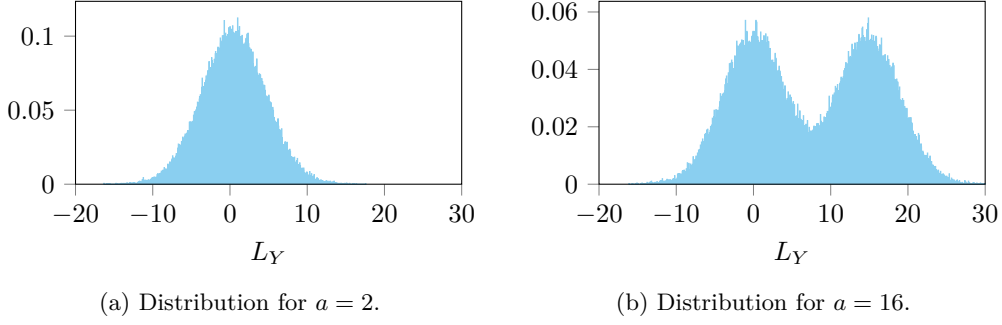


Figure 8: Probability distributions for two different values for the distance of means.

Noise Factor 3. *The distance between leaking sensitive values depends on the practical system it is applied on. A smaller distance leads to a decrease in the noise level.*

5.4 Number of Probing Sets

The number of ways in which an attacker can recover the secret from different sets of leakage values is an additional factor impacting the security level of an implementation. Although the specific relationship between the number of possible attack vectors and the noise level is highly dependent on the algorithm and implementation, it is possible to identify a general guideline for it. In order to derive this, we define a probing set $\mathcal{P} = \{Y_0, Y_1, \dots, Y_V\}$ as a set of V intermediate values that reveals the whole key, i.e., $I(K; \mathcal{P}) = H[K]$. The leakage of the probing set is a vector $\overline{L_Y} = \{L_{Y_0}, L_{Y_1}, \dots, L_{Y_V}\}$. By applying the chain rule of the MI and using the fact that $I(X; Y) \geq 0$ we can conclude that for N different probing sets $\mathcal{P}_i$ it holds

$$I(K; \overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_N}}) \geq I(K; \overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_{N-1}}}).$$

This shows that the MI cannot decrease with more probing sets, i.e., additional observations may add information about the secret but never remove existing knowledge. For a more precise estimate, we assume that each new probing set $\mathcal{P}_j$ contains at least one new Y_x not present in previous sets $\mathcal{P}_i$ with $i, j \in [1, N], i < j$. In particular, since the measurement of the probing set is noisy with σ it holds that $H[\mathcal{P}|\overline{L_{\mathcal{P}}}] > 0$ and $I(K; \overline{L_{\mathcal{P}}}) = I(K; \mathcal{P}) - H[\mathcal{P}|\overline{L_{\mathcal{P}}}] < I(K; \mathcal{P})$. Consequently, if the previous probing sets did not reveal the key completely, an additional probing set $\mathcal{P}_j$ provides the attacker with more information about the secret. Using the entropy, we get

$$H[K|\overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_{N-1}}}] > H[K|\overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_{N-1}}}, \overline{L_{\mathcal{P}_N}}].$$

We can conclude that

$$\begin{aligned} I(K; \overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_N}}) &= I(K; \overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_{N-1}}}) + I(K; \overline{L_{\mathcal{P}_N}} | \overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_{N-1}}}) \\ &> I(K; \overline{L_{\mathcal{P}_0}}, \dots, \overline{L_{\mathcal{P}_{N-1}}}). \end{aligned}$$

Hence, a larger number of probing sets increases the MI between the key and all probing sets, thereby reducing the security level of the implementation. While the exact relation between the noise σ and the number of probing sets N depends on the specific algorithm and, thus, cannot be expressed generically, it can be stated that increasing the number of probing sets leads to an increasing amount of required additional noise. In other words, the more it is computed, the more it is leaked [Sta19]. The practical impact of this factor is demonstrated by the attack of [HNPS24] on the masked FO-comparison of [DBV23].

Noise Factor 4. *A large number of sensitive value combinations that reveal the secret (probing sets) increases the required additional noise.*

6 Practical Application in PQC Masking Schemes

To ensure practical security when designing t -probing secure masking schemes, the above noise factors should be carefully considered. This is because security cannot be guaranteed by the masking order alone; a certain noise-level is also required [BFGV12]. Furthermore, to achieve an exponential increase in security, a linear increase in both the masking order and the noise level is required [DDF14]. In contrast, in our attack on [CGTZ23], the noise level decreases with an increased masking order due to the repetition of sensitive values. This effect, combined with the non-uniformity of the sensitive values makes the implementation more vulnerable to SCAs.

Challenges in PQC. The noise factors are particularly relevant for PQC, as the inherent properties of these schemes often lead to higher noise requirements. Challenges arise, for instance, when PQC schemes require arithmetic computations over larger polynomial rings while embedded devices only involve integer arithmetic and arithmetic over $\mathbb{F}_2$. This results in operations that are modeled as a single instruction in the algorithm’s description, but often require multiple instructions or even an entire subprocedure in practice. Often, this leads to *sensitive values being repeated* (Noise Factor 2) or an increased *distance of means* (Noise Factor 3), thus increasing the noise requirement of the t -probing secure implementation. Interestingly, arithmetic masking over $\mathbb{F}_q$ (where q is prime) provides benefits in both symmetric schemes [MMMS23] and PQC [HNP25]; however under the assumption that uniformity is *always* ensured. Furthermore, applying traditional masking schemes is inefficient for certain procedures. Consequently, more efficient shortcuts are proposed at the cost of introducing non-uniformity (cf. Section 4.6). It should also be noted that, in various PQC schemes, values are distributed non-uniformly by design. As previously derived, the *non-uniformity* (Noise Factor 1) usually increases the amount of noise required to achieve the same practical security level. Another challenge in masking PQC is the vulnerability to distinguishing attacks, where the adversary can distinguish between intermediate values (e.g., [Sta19] and [ABH⁺22]). Side-channel protection is difficult here, because these PQC schemes encode relatively “small” secret information into large data structures. This increases, even for t -probing secure solutions, the *number of probing sets* (Noise Factor 4), and leads to higher noise requirements.

Applicability of Noise Factors to Other (Higher-Noise) Settings. We derived the noise factors from information-theoretic analyses that are target-independent. They give an intuition of how the implementation impacts the noise level, regardless of the absolute noise level. Thus, our analyses and noise factors also apply to other scenarios, like hardware or high-end CPUs, while of course, an inherently higher absolute noise level prevents SCAs. Another fact that must be also considered in these settings is the possibility of parallel computation of sensitive values. This can lead to an amplification of the leakage if a sensitive value is processed n times in parallel. Our analysis for the first noise factor shows that if we consider the n -times processed value as a single sensitive value, the distribution is far away from uniform. So, the noise level of the masking schemes is noticeably decreased.

Arithmetic and Generalized Code-Based Masking. Prime-field masking has been shown to be advantageous over Boolean masking in low-noise settings [MMMS23, HNP25]. While it usually comes with a substantial performance overhead, the increased algebraic complexity may mitigate insufficient noise conditions to some extent. Generalized code-based masking schemes such as direct sum [BCC⁺14, PGS⁺17] and inner-product masking [DF11, BFGV12, BFG15, BFG⁺17], could offer substantial benefits in the practical context [WMCS20, WCG⁺22], but further evaluations, especially regarding the effects on the noise-level, are necessary.

Tool-Based Evaluation. To the best of our knowledge, there are currently no tools that automatically analyze a masking scheme during implementation with respect to the required noise level. Nonetheless, such analysis should be feasible, and the verification tool QUANTILE [HCP⁺24] represents a step in this direction by computing bounds on the MI. It is also possible to automatically verify the derived noise factors during implementation, but they should already be considered in the design phase of the masking scheme. In order to support designers in this early stage, we propose the following practical guidelines.

Practical Guidelines. Our attack on the *t*-probing secure masking scheme on FWPS of [CGTZ23] or the attack of [HNPS24] on the masked FO-comparison implementation of [DBV23], show that the noise requirements must be carefully considered when designing and implementing a theoretically *t*-probing secure masking scheme. Rather than analyzing the MI at every point of the implementation (a particularly difficult task in PQC), the following guidelines and design strategies help to avoid unexpectedly high noise requirements:

- (i) Ensure that the leakage values L_Y are distributed uniformly or, even better, binomially. This can be achieved by ensuring that the sensitive values Y (i.e., the shares) are distributed uniformly. A potential design strategy to address this issue is to ensure that, on a 32-bit system, the magnitude of the value range is 2^{32} .
- (ii) Always consider the instruction set – and its impact on the highlighted factors – already when designing the masking scheme. For example, abstractions beyond the granularity of the instruction set should be avoided, and any requirements for specific instructions on complex operations (e.g., permutations) should be explicitly outlined.
- (iii) Identify all sensitive values and encode them in as few bits as possible in order to avoid increasing the distance of means. This also decreases the number of probing sets, if the encoded bits are processed sequentially. For instance, when storing a conditional swap flag or sign bit, encode it as a single bit rather than a full word.
- (iv) If the masking scheme increases the required additional noise by any of the previously derived factors, introduce the necessary noise into the system using hiding techniques. For instance, this could be achieved by adding a noise generator like [LCL10] to the system or by randomizing the computation as in [LBB19]. Hiding helps to close the gap, as this artificial noise adds to the system-inherent noise, thus increasing the overall noise level [CCD00]. In most cases, a combination of hiding and masking is the most efficient way to counteract SCAs [LBBS20]. However, hiding techniques may be inefficient and complex, as generating noise that hides the sensitive signal exactly, without consuming an unnecessarily high amount of power, is not trivial. Further research is therefore needed in this area.

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